

Lab 3: Confounding & Effect Measure Collapsibility

Overview

In this lab you will explore confounding and effect measure collapsibility using the Right Heart Catheterization (RHC) dataset, investigating the association between receiving a right-heart catheterization and all-cause mortality. You will prepare the data, produce a descriptive summary table, and fit a series of regression models to calculate and compare crude, conditional, and marginal effect estimates (odds ratios, risk ratios, and risk differences) — seeing how adjustment affects each measure differently. Your TA will walk through the worked solution during the session.

This is the workflow you will reuse in **P2** on your own locked paper.

Materials

- **Exercise instructions:** Confounding
- **Worked solution:** The R codes

Before you come to lab, install the package every model chunk uses:

```
install.packages("Publish")    # capital P; the function you call is lowercase publish()
```

Three things to know about the downloaded exercise file:

- **Uncomment the read.csv line** in the first chunk. The data are in the zip's Data/ folder, and nothing else runs until `df` exists.
- It says the RHC data were used in “Lab Assignment #1”. That was another course. The wrangling code is given in full, and you are not expected to have seen these data.
- Ignore its percentage grade weights. This lab is graded Complete / Incomplete (below).

Your task

1. Follow along as the TA explains the worked solution.
2. Run the R Markdown file yourself, as a group, on your own laptop.
3. Put your **group name and the first names of everyone present** in the `author:` field at the top of the file.
4. Add `sessionInfo()` in a final code chunk.
5. Knit to **PDF or HTML** and submit that one file before the lab ends.

No written explanations are required. If you get stuck, ask the TA in the room — that is what the session is for.

Deliverable

- **One file per group:** the knitted PDF or HTML.
- **Due:** end of the lab session (Thu Oct 1).

Grading: Complete / Incomplete

You receive **Complete** if the document knitted without errors, your group name and members are in it, and it was submitted before the lab ended. Only members present in the room receive credit.

Where this lab fits

This lab is one step in a weekly chain: Tuesday's lecture sets up the method, Thursday's lab runs it on fixed teaching data, and your increment applies it to the paper your group locked at M0.

	Document	How this lab connects
This week's lecture	Week 4 — Confounding and DAGs; Table 1; variable roles	Tuesday used the MASLD paper to decide what to adjust for — DAGs, the empirical criteria, why change-in-estimate and stepwise selection fail — and showed why an odds or hazard ratio moves on adjustment even without confounding. This lab runs the mechanics on a different dataset (RHC): crude, conditional and marginal odds ratios, risk ratios and risk differences, and standardisation.
Slide deck	Week 4 deck	The same material as slides, with speaker notes. Press S for notes, F for fullscreen.
Your increment	P2 — Confounding diagnostic	Produce your own Table 1 plus a written adjustment rationale — which variables you adjust for, which you deliberately do not, and why.
Worked example	P2 example — its code	A completed P2: the Table 1, and the adjustment rationale written out. Read the rationale first — it is the part that carries the marks.
Milestone	M1 — Replication and interrogation	P2 enters M1 as Confounding, adjustment set, Table 1 . P2 is optional and ungraded with nothing to submit; M1 is where it is assessed.